MA2101 :Analysis I

Maximum Marks: 50

Problem 1.
(a) Let $\{x_n\}$, $\{y_n\}$ and $\{z_n\}$ be three sequences of real numbers. Show that if $x_n \leq y_n \leq z_n$ for all $n \in \mathbb{N}$, and if $\lim x_n = \lim z_n = l$, then $\lim y_n = l$.

(b) Let $\{x_n\}$ and y_n be two sequences of real numbers. Define z_n to be the shuffled sequence $\{x_1, y_1, x_2, y_2, x_3, y_3, ..., x_n, y_n, ...\}$. Show that z_n is convergent if and only if $\{x_n\}$, $\{y_n\}$ are both convergent with $\lim x_n = \lim y_n$.

(c) Show that the limit of the sequence $b_n = n - \sqrt{n^2 + 2n}$ exists and find its limit.[3+4+3=10]

Problem 2.
(a) Consider the series $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^p log(n)}$. Find all p for which the series is (i) conditionally convergent and (ii) absolutely convergent. [10]

(b) Consider the alternating series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{8} + \frac{1}{6} - \frac{1}{16} + \frac{1}{7} - \frac{1}{32} + \frac{1}{8} - \ldots$. Argue whether the series is conditionally or absolutely convergent. [4]

Problem 3.
(a) Define the cantor set. Show that the cantor set is perfect i.e. it is closed and it contains no isolated points.

(b) A set $E \in \mathbb{R}$ is totally disconnected if for any two distinct points $x, y \in E$, there exist separated sets A, B such that $x \in A$, $y \in B$ and $E = A \cup B$. Show that the cantor set is totally disconnected. [5+5=10]

Problem 4.
(a) A set $G \subset \mathbb{R}$ is dense in $\mathbb{R}$ if, given any two real numbers $a < b$, it is possible to find a point $x \in G$ with $a < x < b$. If $\{G_1, G_2, G_3 \ldots\}$ is a countable collection of dense, open sets, then show that the intersection $\cap_{n=1}^{\infty} G_n$ is non-empty.

(b) Let K, L be two disjoint compact subsets of the real numbers. Define $d := \inf\{|x - y| : x \in K,\ y \in L\}$. Show that $d > 0$ and there points $x_0 \in K$ and $y_0 \in L$ such that $d = |x_0 - y_0|$. [6+5=11]

Problem 5.
Suppose $\{x_n\}$ is a monotonically increasing sequence which is also bounded above. Then show that the sequence $\{x_n\}$ converges to $\text{Sup}\{x_n : n \in \mathbb{N}\}$. [5]